



Multi-Agent Student Calendar Planner

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GROUP 45

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Introduction

Students often manage several deadlines, grade weights, uncertain task durations, and personal constraints at the same time.

Our project treats student planning as a multi-agent coordination problem:

- Different agents defend different priorities;
- Agents can disagree about the same calendar;
- The final plan emerges from critique, revision, and quorum approval.

Related Work: MACI

Our main related work is MACI: Multi-Agent Collaborative Intelligence for Adaptive Reasoning and Temporal Planning.

MACI proposes a multi-agent planning process where one component generates a plan, while other agents independently validate it and point out problems. The plan is then refined through several iterations. We are not reproducing the full MACI framework. Instead, we reproduce its core idea in a simpler domain:

Planner creates a calendar

→ Multiple specialist agents critique it

→ Planner revises the calendar

<https://arxiv.org/abs/2501.16689>

Problem Definition & Objectives

Problem:

Given a natural-language description of a student's tasks, deadlines, availability, sleep/energy state, and constraints, generate a feasible multi-day calendar. The output should not only say what to do. It should also explain why each compromise was made.

Objectives:

1. interpret natural-language task descriptions using the current date;
2. represent competing planning priorities through specialist agents;
3. generate calendars with explicit compromise reasoning;
4. revise calendars based on agent critiques;
5. accept a calendar only when quorum and hard constraints are satisfied.

Approach

Inputs:

- user natural-language task description;
- computer current date/time;
- required approvals, quorum Q.

Output:

- final calendar;
- stop reason;
- agent approvals;
- remaining concerns and assumptions.

Environment state:

- interpreted tasks, deadlines, availability, and uncertainties;
- agent perspective memory;
- calendar version history;
- critique history;
- constraint validation results.

Work Type: Agent Orchestration

Our focus is coordinating multiple agents. We are not training them, but studying how several agents can coordinate around a shared calendar plan.

We will analyze:

- How agents with different priorities disagree and reach agreement;
- How the quorum value affects strictness and convergence;
- How many critique/revision rounds are needed;
- Computational cost in terms of LLM calls: agents × iterations × scenarios.

Relevant methods:

We use a simple stateful workflow with reflection: the system moves through interpretation, planning, critique, validation, and finalization, with agents critiquing the plan before it is revised.

For now, RAG and long-term memory are not central now, but could be useful later if we integrate real calendars or user history.

System Architecture

1. The User writes the tasks in natural language and chooses the quorum value Q .
2. The Interpreter Agent extracts tasks, deadlines, availability clues, and uncertainties.
3. The Specialist Agents analyze the same information from different perspectives:
 - Deadline Agent
 - Grade Agent
 - Effort Agent
 - Wellbeing Agent
 - Risk Agent
4. The Planner–Arbiter creates a calendar version and includes the reasons behind the compromises.
5. The Specialist Agents critique that same calendar.
6. The Constraint Checker validates hard rules, such as deadlines and available time.
7. The Planner–Arbiter revises the calendar if the acceptance criteria are not met; otherwise, the calendar is finalized.

Evaluation

For the first version, we are not modelling real calendar apps, travel time, notifications, live rescheduling, group work negotiation, or long-term learning from past plans.

We chose quorum with critical critiques because it is simple to understand: the plan is accepted only if enough agents approve and no agent finds a serious problem. If no plan is fully accepted after 3 iterations, we fall back to the best calendar version found so far.

There is a trade-off between precision and efficiency: more agents and more revision rounds can improve checking, but also increase LLM calls and latency. To keep it practical, we limit the process to 3 iterations.

We do not fully evaluate long-term student satisfaction, robustness to very vague input, or performance against real calendar tools.

Evaluation

Each agent protects a different priority: deadlines, grade value, effort realism, wellbeing, or risk.

The agents work on one shared candidate calendar. They critique the same candidate calendar and negotiate through the Planner–Arbiter.

Consistency is maintained because every critique refers to the same calendar version, and the Planner–Arbiter tracks what changed between revisions.

A plan is accepted only when hard constraints pass, no critical critique remains, and the required quorum is reached.